BRICK WORK · MILESTONE 4.0

Data model 4.0

Goal: Brick Work runs on schema 4.0 in prod. Every set, part, minifigure and colour has a Brick Work id; the catalogue comes from Rebrickable’s dump with no key; the collection is migrated from 3.3 with a proof; API v2 replaces v1.

Rough length: 18–22 weeks · Before milestone A · Gate: issue #116 · ADR 0016

WHY

BrickLink numbers cannot carry a catalogue that comes from Rebrickable

Today, 0.5

BrickLink’s type and number is the identity of everything. What cannot be translated becomes a “?” item. Sticker sheets, legs whose BrickLink id depends on colour, and assemblies have no place.

After 4.0

Brick Work gives every item its own id. BrickLink, Rebrickable and Brickset link to it with a confidence: matched, inferred, manual or to check, scoped to a colour where needed.

Spike #102 settled the source: Rebrickable’s dump is free to use with acknowledgement, BrickStore’s archive of BrickLink data has no shown permission. The identity touches 54 key references in 40 tables, so the whole schema is re-derived.

WHAT CHANGES FOR THE USER

A reference book and a notebook

Area	In 4.0
Two files	catalogue.db is rebuilt from the dump; brickwork.db holds the collection and is never touched by a rebuild
Adding a set	Offline and instant from the catalogue; a part-list picker when there is a choice
Part lists	Rebrickable by default; BrickLink’s list per set or by a global switch; “version n of m”
Keys	None needed to start; BrickLink adds prices, order import and its part lists
Migration	Collection, prices, keys and tokens, preferences; history stays in the old file
API	v2 by Brick Work id with number lookups; v1 answers 410; Toni and lego-add switch

THE WORK

Twelve pieces, about 20 weeks

Item	Piece	Weeks
M.1	Schema and connection	1
M.2	Catalogue builder	2.5
M.3	Shipped mappings	1
M.4	Figure mapping test	0.75
M.5	Acquired part lists	0.75
M.6	Copies and switching	1

Item	Piece	Weeks
M.7	Migration with proof	2
M.8	Core modules on item_id	4.5
M.9	Test-suite rewrite	2.5
M.10	Screens, frames first	2
M.11	API v2 and clients	1.5
M.12	Onboarding, checks, cutover	0.75

Plus 1 week of contingency. M.1, M.3 and M.4 can start together. The least certain lines are the core modules, the test rewrite and BrickStore’s file format.

THE GATE · #116

Prod runs on 4.0, proven against 3.3

- M.1–M.12 closed
- Migration proof PASS on a copy of prod on test, rehearsed with the cutover catalogue
- Cutover decided (#115), prod on 4.0, the 3.3 file kept beside it
- Toni and lego-add working against API v2
- Catalogue file in Borg, first archive checked
- Close-out: this deck becomes the milestone’s status report